Lecture 9: Bay's Theorem

29 June 2026 12:40

<u>Topics of this lecture:</u>

1. Bay's Theorem

According to conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

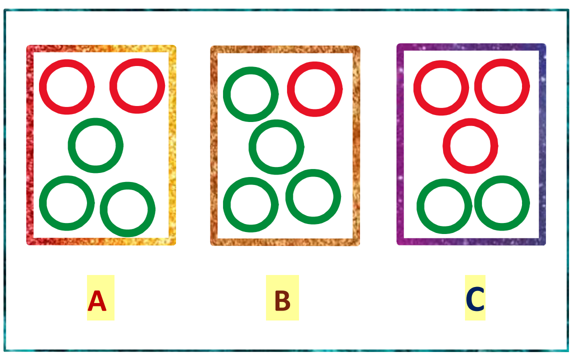
According to multiplication theorem:

$P(A \cap B) = P(B \mid A) \,.\, P(A)$

Now we can write:

$$P(A|B) = \frac{P(B \mid A) \,.\, P(A)}{P(B)}$$

This is the expression of Bay's Theorem



Q. Pick a bag at random
And draw a random ball out of it.

1. **Probability of getting a red ball given bag a is chosen ?**

   Conditional Probability
   $P(\bigcirc \mid A) = 2/5$

2. **Probability of getting a red ball and choosing bag A ?**

   Multiplication Theorem
   $P(\bigcirc \cap A) = P(\bigcirc \mid A) \,.\, P(A)$
   $= 2/5 \,.\, 1/3$
   $= 2/15$

3. Probability of getting a red ball ?

Total Probability

P(○) = P(○ | A) . P(A) + P(○ | B) . P(B) + P(○ | C) . P(C)

= 2/5 . 1/3 + 1/5 . 1/3 + 3/5 . 1/3

= 2/5

Total Probability

P(A) = 1/3 P(A) = 1/3 P(A) = 1/3

P(○ | A) = 2/5 P(○ | B) = 1/5 P(○ | C) = 3/5

4. Probability of bag a is chosen given that the drawn ball is red ?

Total Probability

$$P(○) = \underbrace{P(○ \mid A)\ .\ P(A)}_{\text{Contribution of Bag A}} + \underbrace{P(○ \mid B)\ .\ P(B)}_{\text{Contribution of Bag B}} + \underbrace{P(○ \mid C)\ .\ P(C)}_{\text{Contribution of Bag C}}$$

$$P(A \mid ○) = \frac{P(○ \cap A)}{P(○)} = \frac{\text{Contribution of Bag A}}{\text{Total Probability of red ball}}$$

$$P(A \mid ○) = \frac{2/5\ .\ 1/3}{2/5\ .\ 1/3 + 1/5\ .\ 1/3 + 3/5\ .\ 1/3}$$

$$P(A \mid ○) = \frac{2/15}{2/5} = 1/3$$

2/5

Q2. Suppose two factories supply light bulbs to the market

- Factory A's bulb work for over 5000 hours 99% of the cases.
- Factory B's bulb work for over 5000 hours 95% of the cases.

Factory A supplies 60% of the total bulbs.
What is the chance that a randomly purchased bulb will work for more than 5000 hours ?

Let event E = randomly purchased bulb will work for more than 5000 hours

Total Probability

P(E) = P(E | A) . P(E) + P(E | B) . P(E)

P(E) = 0.99 . 0.6 + 0.95 . 0.4

P(E) = 0.97

Given that -
The purchased bulb worked for more than 5000 hours.
What is the probability that it was produced by factory A?

$$P(A \mid E) = \frac{\text{Contribution of Factory A}}{\text{Total Probability of E}} = \frac{P(E \mid A) . P(E)}{P(E \mid A) . P(E) + P(E \mid B) . P(E)}$$

P(A | E) = (0.99 X 0.6) / (0.97)

= 0.6124

probability that it was produced by factory A is 61 %